Program overview

08-Dec-2022 11:29

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor MT 2022			
1e jaar BSc MT 2022	**1e Year BSc MT 2022**		
Wiskunde blok			
WBMT1050	Mathematics 1	6	
WBMT1050 Toets 1	*Analyse 1 - deeltentamen*	*3*	
WBMT1050 Toets 2	*Analyse 2 - deeltentamen*	*3*	
WBMT1051	Mathematics 2	6	
WBMT1051 Toets 1	*Lineaire Algebra 1 - deeltentamen*	*3*	
WBMT1051 Toets 2	*Lineaire Algebra 2 - deeltentamen*	*3*	
Natuurkunde blok			
WB1135	Introductory Dynamics	6	
WB1530-14	Thermodynamics	6	
WB1630-20	Statics	6	
WB1631-15	Mechanics of Materials	6	
Maritieme Techniek blok			
MT1455	Geometry and Stability	6	
MT1456	Constructing	6	
MT1457	Resistance, Propulsion & Drive Systems	6	
MT1458	Integration Project	6	
2e jaar BSc MT 2022	**2nd Year BSc MT 2022**		
MT201-11 Wiskunde blok	**Mathematics block**		
WBMT2048	Mathematics 3	6	
WBMT2048 Toets 1	*Analyse - deeltentamen*	*3*	
WBMT2048 Toets 2	*Differentiaalvergelijkingen - deeltentamen*	*3*	
WBMT2049	Mathematics 4	6	
WBMT2049 T1	*Kansrekening en Statistiek - deeltentamen*	*3*	
WBMT2049 T2	*Numerieke Wiskunde - deeltentamen*	*3*	
WBMT2049 T3	*Numerieke Wiskunde - practicum*	*0*	
MT203-11 Maritieme Theorie blok	**Maritieme Theorie blok**		
MT2430-16	Weerstand, Voortstuwing en Aandrijving 2	6	
MT2430-16 Toets 1	*WVA2 Tussentoets*	*1*	
MT2430-16 Toets 2	*Voorstuwingspracticum*	*0,5*	
MT2430-16 Toets 3	*Cavitatiepracticum*	*0,5*	
MT2430-16 Toets 4	*Dieselmotorenpracticum*	*0,5*	
MT2430-16 Toets 5	*WVA2 tentamen*	*3,5*	
MT2432-22	Strength of Ships	6	
MT2433	Scheepsbewegingen	6	
WB2630	Advanced Mechanics	6	
WB2630 Toets 1	*Rigid-Body Dynamics*	*3*	
WB2630 Toets 2	*Continuum Mechanics*	*3*	
Maritieme Praktijk blok			
MT2429-22	Maritime Markets and Operations	6	
MT2431	Geometrie en Stabiliteit 2	6	
MT2431 Toets 1	*Project Opdracht*	*3*	
MT2431 Toets 2	*G&S2 Tentamen*	*3*	
MT204-11 Maritieme Project blok	**Maritieme Project blok**		
MT2434	2e Integratie Project	12	
MT2434 Toets 1	*Project Opdracht(en)*	*10*	
MT2434 Toets 2	*Ethiek en Veiligheid tentamen*	*2*	
3e jaar BSc MT 2022	**3rd Year BSc MT 2022**		
3e jaar BSc MT minor 2022	**3rd Year BSc MT Minor 2022**		
3e jaar BSc MT major 2022	**3rd Year MSc MT Major 2022**		
MT3420	Elektrische Systemen & Regeltechniek	6	
MT3432	Organisatie van Scheepsproductie	6	
MT3432 Toets 1	*OVS Tentamen*	*2,5*	
MT3432 Toets 2	*Report*	*3,5*	
MT3433	Complexe Belastingen en Trillingen	6	
MT3BEP	Bachelor End Project	12	

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Bachelor MT 2022

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1e jaar BSc MT 2022

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Wiskunde blok

WBMT1050	**Mathematics 1**	**6**
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	

WBMT1050 Toets 1	***Analyse 1 - deeltentamen***	***3***
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	1	
Start Education	1	
Exam Period	1 2	

WBMT1050 Toets 2	***Analyse 2 - deeltentamen***	***3***
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	2	
Start Education	2	
Exam Period	2 3	

WBMT1051	**Mathematics 2**	**6**
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	Dutch	

WBMT1051 Toets 1	***Lineaire Algebra 1 - deeltentamen***	***3***
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3	
Start Education	3	
Exam Period	3 4	

WBMT1051 Toets 2	***Lineaire Algebra 2 - deeltentamen***	***3***
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	4 5	

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Natuurkunde blok

WB1135	Introductory Dynamics	6
Responsible Instructor	Prof.dr. P.G. Steeneken	
Instructor	R.F. Dooren	
Instructor	F.G.J. Broeren	
Instructor	Ir. A.M. van der Niet	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering 3mE Department Cognitive Robotics	

WB1530-14	Thermodynamics	6
Responsible Instructor	Dr. R. Pecnik	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB1630-20	Statics	6
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Course Coordinator	F.G.J. Broeren	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1631-15	Mechanics of Materials	6
Responsible Instructor	Dr.ir. M. Langelaar	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

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Maritieme Techniek blok

MT1455	Geometry and Stability	6
Responsible Instructor	Dr.ing. C.H. Thill	
Instructor	R.H. van Till	
Instructor	E.H.M. Ulijn	
Instructor	Ir. A.A. Nijdam	
Instructor	Ing. J. Rodrigues Monteiro	
Instructor	Dr.ing. S. Schreier	
Instructor	Ing. B. Goris	
Instructor	M.J.G. van der Drift	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Materials Science & Engineering	

MT1456	Constructing	6
Responsible Instructor	Dr.ir. J.F.J. Pruijn	
Instructor	Ir. J.J.B. Teuben	
Instructor	Dr.ir. J.H. den Besten	
Instructor	Ing. B. Goris	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1457	Resistance, Propulsion & Drive Systems	6
Responsible Instructor	E.H.M. Ulijn	
Instructor	A. Glasbergen-Plas	
Co-responsible for assignments	R.H. van Till	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1458	Integration Project	6
Responsible Instructor	Ing. B. Goris	
Responsible Instructor	Dr. A.A. Kana	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

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2e jaar BSc MT 2022

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MT201-11 Wiskunde blok

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics,	
Expected prior knowledge	First year Mathematics courses	
Course Contents	WBMT2048/T1: Series and sequences. Line and surface integrals. Integral Theorems of Green, Stokes and Gauss. WBMT2048/T2: Differential Equations (linear first and second order equations, Laplace transform, systems of linear equations, autonomous systems and stability, Fourier series, partial differential equations)	
Study Goals	Study goals for each part can be found on the corresponding CourseBase-page (WBMT2048/T1 or WBMT2048/T2).	
Education Method	Online Lectures, 4 hours per week.	
Literature and Study Materials	WBMT2048/T1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace. WBMT2048/T2: W.E. Boyce, R.C. DiPrima, D.B. Meade, Boyce's Elementary Differential Equations and Boundary Value Problems, global edition, 2017.	
Assessment	WBMT2048/T1: On-campus exams. Partly digital (in Grasple), partly written exams (see Brightspace for more details). There will be two partial exams: one partial exam in week 5 about the first part of the course (Sequences and Series) and one partial exam in week 10 about the second part of the course (Vector Calculus). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, about the complete course, that lasts 180 minutes. It is not possible to resit only one partial exam. If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace). WBMT2048/T2: On-campus exams. Partly digital (met Grasple), partly written (see Brightspace for more details). If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace).	
Exam Hours	WBMT2048/T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048/T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048/T1: A formula sheet and a flow chart series (see Brightspace). NO calculators. WBMT2048/T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	

WBMT2048 Toets 1	*Analyse - deeltentamen*	*3*
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics	
Expected prior knowledge	First year Mathematics courses	
Course Contents	Series and sequences. Line and surface integrals. Integral Theorems of Green, Stokes and Gauss.	
Study Goals	After this course the student should be able to 1. determine whether a sequence is convergent or divergent, for example by calculating its limit; 2. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 3. determine the interval of convergence of a power series; 4. determine the Taylor series of a simple analytic function, for example sin(x), exp(x), ln(x), arctan(x), (1+x)^a; 5. find parametrizations of curves and surfaces (in 2D and 3D); 6. calculate a line integral or a surface integral of a function or a vector field; 7. mention the physical interpretation of these line and surface integrals; 8. define and formulate the physical meaning of concepts like gradient, curl and divergence; 9. calculate line and surface integrals via one of the three integral theorems (Green, Stokes and Gauss).	
Education Method	Lectures, 4 hours per week (2x2).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital (in Grasple), partly written exams (see Brightspace for more details). There will be two partial exams: one partial exam in week 5 about the first part of the course (Sequences and Series) and one partial exam in week 10 about the second part of the course (Vector Calculus). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, about the complete course, that lasts 180 minutes. It is not possible to resit only one partial exam. If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace).	
Exam Hours	Each partial exam: 90 minutes. The resit: 180 minutes.	
Permitted Materials during Tests	A formula sheet and a flow chart series (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Remarks	This is half of the course WBMT2048. Partial grades for T1 and T2 will be given in one decimal. The final grade is a number rounded to the nearest half. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So the average is 6,4. The final grade will be: 6,5. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75).	
Judgement	See Dutch comment.	

WBMT2048 Toets 2	*Differentiaalvergelijkingen - deeltentamen*	*3*
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

WBMT2049	**Mathematics 4**	**6**
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3 4	
Start Education	3 4	
Exam Period	Different, to be announced	
Course Language	Dutch	

WBMT2049 T1	*Kansrekening en Statistiek - deeltentamen*	3
Responsible Instructor	Dr.ir. R. Versendaal	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Required for	Bachelor project, multivariate data analysis	
Expected prior knowledge	Techniques from first year math: integration, differentiation, the exponential function and the natural logarithm. Analyse 1, 2 and 3, Linear Algebra.	
Parts	The course is split in two parts: In the first part the basis of probability theory is presented with important distributions (such as uniform, normal, exponential, binomial, Bernoulli, geometric), random variables and their expectation and variance. Furthermore, joint distributions and their properties (such a covariance and correlation) are treated. In the second part it is explained how to summarize and interpret datasets, both graphically and numerically. Next, confidence intervals and hypothesis testing are treated.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebyshev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Lectures with interactive elements; pre-lecture videos, online exercises	
Computer Use	Python	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Prerequisites	Analysis 1, 2, 3, linear algebra	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one final exam (week 10) on Part 2: Statistics. Final grade is the average of both tests.	
Exam Hours	90 min	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Simple (non-graphical and non-programmable) calculator	

WBMT2049 T2	*Numerieke Wiskunde - deeltentamen*	3
Responsible Instructor	Dr. N.V. Budko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

WBMT2049 T3	Numerieke Wiskunde - practicum	0
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See WBMT2049, Numerical mathematics part	
Study Goals	See WBMT2049, Numerical mathematics part	
Education Method	Practicum	
Practical Guide	One Jupyter Notebook per assignment must be completed and submitted. Notebooks can be submitted by a group of at most two students: these students must register as group in Brightspace (see above). Only students in a registered group are allowed to cooperate on the assignment. Notebooks can only be handed in through Assignments within Brightspace. Any notebook handed in any other way will not be assessed. Distributing to others and/or receiving from others the assignment and/or the resulting notebook is prohibited. With others we mean any party not being your group partner, the lecturers and/or the assistants. An action of one student counts as an action of the entire registered group. If any of the above rules are violated by a group, any assessment of the Notebooks of the violating group will be revoked.	
Assessment	On basis of a submitted Jupyter Notebook, resulting in either a V (pass) or an O (fail).	
Enrolment / Application	A separate electronic subscription for the numerical practicum might (also) required, see Brightspace for details. On top of that for participation in the course Mathematics 4 an early subscription for the module Mathematics 4 is required as well. More information regarding the general rules can be found at http://onderwijs.3me.tudelft.nl/osiris.	
Remarks	It concerns a partial score. Needed is a completion (V).	
Judgement	For rules concerning education you may consult http://onderwijs.3me.tudelft.nl/reglementen.	

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MT203-11 Maritieme Theorie blok

MT2430-16	Weerstand, Voortstuwing en Aandrijving 2	6
Responsible Instructor	Ir. K. Visser	
Instructor	Dr.ir. A. Vrijdag	
Instructor	Prof.dr.ir. T.J.C. van Terwisga	
Instructor	Dr.ing. C.H. Thill	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Study Goals	After successfully completing this course the student is able to: With respect to: - Resistance: Describe ITTC 72 breakdown of ship resistance into components Explain (and apply some) methods for resistance prediction: model tests, systematic series, empirical prediction programs, RANS - Propulsion: Describe and apply actuator disk theory and Open Water diagrams Explain lifting line theory and propeller design Explain the relation between propeller geometry and performance and cavitation Explain cavitation, erosion and pressure fluctuations Calculate Thrust, Torque, Power, Efficiency, etc. of propellers - Resistance and Propulsion (R&P): Derive and explain properties of fluid dynamic models in R&P: RANS, Euler, Laplace, thin boundary layer equations Describe Propeller-hull interaction Explain application of BEM (panel methods) and RANS methods in R&P Describe propeller-hull optimization in ship design and Energy Saving Devices - Driving components (prime movers like diesel engine or gas turbine): Explain and apply torque-speed/power-speed characteristics, construction types, ideal process models and actual processes, turbo-charging concept Calculate Torque, Power, Efficiency, Mean Effective Pressure, fuel and air consumption, air excess ratio and load related units (in thermodynamical terms) of engines - Propulsion systems (combination of drive + propulsion components to overcome resistance): Explain different lay-outs of propulsion systems for different ships (applications) Match propulsor to ship and prime mover to propulsor Analyse (off-)design behaviour of propulsion systems - Pump pipe flow systems: Explain different lay-outs of pump pipe flow systems for different applications (in ships) Analyse stationary behaviour of pump pipe flow systems in Q-H diagram Calculate pressures, (flow) velocities, power, etc. - Emissions: Describe different emissions from ships (to air and water) Explain the root of harmful emissions from chemical composition / processes Describe regulations to limit emissions from ships and their effects Describe possible counter measures / emission abatement systems Course material underlying theory: - Conservation laws from physics (conservation of mass, momentum and energy). - Fluid mechanics (describing boundary layer, continuity, properties of material flow (liquid), Bernoullis equation). - Thermodynamics (ideal gas, cyclic processes, properties of material flow (gaseous), second law). - Chemistry (high school chemistry is sufficient)	
Department	3mE Department Maritime & Transport Technology	

MT2430-16 Toets 1	*WVA2 Tussentoets*	*1*

MT2430-16 Toets 2	*Voorstuwingspracticum*	*.5*

MT2430-16 Toets 3	*Cavitatiepracticum*	*.5*

MT2430-16 Toets 4	*Dieselmotorenpracticum*	*.5*

MT2430-16 Toets 5	*WVA2 tentamen*	*3.5*

MT2432-22	Strength of Ships	6
Responsible Instructor	Dr. C.L. Walters	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Description box (5 ECTS): The course consists of four parts: - Strength of ships: (1) lectures + (2) laboratory exercise - Material science: (3) lectures + (4) laboratory exercise (1) Strength of ships lectures Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.: a) Stress concentration factors b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson; c) Buckling of plates: loading, boundary conditions, geometries; d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading; e) Plasticity: form factors, maximum capacity of beams, hinge line theory f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory. The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended: - Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library)(2) Strength of ships laboratory exercise: plate test (3) Material sciences lectures a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties). b) Strengthening mechanisms on the microstructural scale. c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations). d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability. e) Most common welding processes, process descriptions, advantages and disadvantages. f) Influence of welding on materials microstructure, residual stress, deformations, common defects. (4) Material sciences laboratory exercise	
Study Goals	(1) + (2) Strength of ships The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads. The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation). (3) + (4) Materials sciences Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures. The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection. The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.	
Education Method	Lectures and laboratory exercises	
Assessment	a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5ECTS	
Enrolment / Application	To participate in this course, the students must sign up on time. More information about signing up for education, admission requirements, and current deadlines can be found on: https://www.tudelft.nl/index.php?id=50357&L=1 Signing up for exams needs to be done separately. *in some situations, there are prerequisites for a course. These can be found in the following rules: https://www.tudelft.nl/index.php?id=33610&L=1	
Department	3mE Department Maritime & Transport Technology	
Judgement	For the regulations (e.g. signing up for education and validity of grades) please refer to the Educational regulations, http://onderwijs.3me.tudelft.nl/reglementen	

MT2433	Scheepsbewegingen	6
Responsible Instructor	Dr.ir. H.J. de Koning Gans	
Instructor	Dr.ir. H.J. de Koning Gans	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	The course lectures the basic knowledge and theory converning shipmotions in waves. The course addresses: * Wave theory in deep and shallow water * Calculation methods based on potential flow theory, like 2D strip-theory * Equations of motions for a ship in waves * Lifting theory applicable to rudders and fins * Experimental techniques to establish ship motions (the towing tank experiments)	
Study Goals	After the course the student is able to: * Calculate pressures and velocities in regular waves * Calculate local motions and accelerations on board a vessel * Explain the concept of transfer functions for vessel motions * Explain the forces on lifting surfaces like rudders As well, the student is able to: * Explain the motion equations for a ship and explain the theory of ship motions * Describe the parameters in the motion equations and how they can be established by means of experimental techniques	
Education Method	Lectures and Workshops	
Literature and Study Materials	Lecture material will be posted on Brightspace	
Assessment	Written examen	
Remarks	It is essential that the following course have been followed and that the material is 'known' at the start of the lectures. This concerns: "Scheepshydrostatica, Analyse, Lineaire algebra ,Differentiaalvergelijkingen, Rigid body dynamics and Mechanics".	
Department	3mE Department Maritime & Transport Technology	

WB2630	Advanced Mechanics	6
Responsible Instructor	Prof.dr. H. Vallery	
Responsible Instructor	Dr.ir. L.F.P. Noel	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	3mE Department Biomechanical Engineering 3mE Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

Responsible Instructor	Prof.dr. H. Vallery
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hourse 2/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	Dutch
Expected prior knowledge	Prerequisites for the course: see document on Brightspace or navigate to dbl.tudelft.nl-> Student Resources->preparatory material for the course "Rigid-Body Dynamics" Note that the first test will also cover prerequisites.
Course Contents	1) Kinematics of a rigid body in 3D a. Description of single and successive 3D rotations via Euler angles, rotation matrices, axis and angle of rotation. b. Angular velocity and instantaneous axis of rotation c. Transport Theorem 2) Kinetics of a rigid body in 3D a. The inertia tensor b. Principle of work & energy c. Newton-Euler equations d. Free rotational motion, gyroscopic effects 3) Systems with n degrees of freedom a. Generalized coordinates b. Principle of virtual work c. Lagranges equations 4) Methods to check plausibility of generated results, such as a. Units notation and checks b. Numerical approximation c. Checking validity of assumptions d. Treatment of special cases
Study Goals	Kinematics of a rigid body in 3D The student is able to describe rotations of a rigid body in 3D via Euler angles, rotation matrices, and the principal axis of rotation. transform one description of rotations into the other, and re-write successive rotations as a single rotation in the different representations. describe angular velocity of a rigid body in 3D and find the instantaneous axis of rotation Kinetics of a rigid body in 3D The student is able to calculate the inertia tensor for a rigid body via integration analyze the inertia tensor and find the principal axes derive the equations of motion for a rigid body in 3D using the principle of work and energy or using the Newton-Euler equations recognize and explain gyroscopic effects Systems with n degrees of freedom The student is able to describe general motion of systems of particles and of rigid bodies in generalized coordinates in 2D and 3D apply the principle of virtual work to derive the equations of motion for systems of particles and rigid bodies in 2D and 3D formulate kinetic and potential energy for systems of particles and for rigid bodies in generalized coordinates apply Lagranges methods to systems of particles and to rigid bodies to obtain the equations of motion Recognize conservative systems Generating reliable results to mechanical problems The student is able to test the plausibility of calculation steps and final results be aware of and check validity of assumptions whenever applying learned concepts
Education Method	Students prepare the course material themselves each week. For this, they can use the book and material provided on Brightspace. On Brightspace, also an indication of the weekly topics will be provided. Each week, interactive on-campus lectures and on-campus tutored sessions are organized. During the tutored sessions, students can work on homework exercises and ask for further explanation and help. The first week, a formative digital test is provided to help students understand their current level with respect to prerequisities. In contrast to the mid-term and the final exam (see below), this test does not count towards the courses grade.
Books	Prerequisites for the course: see document on Brightspace or navigate to dbl.tudelft.nl-> Student Resources->preparatory material for the course "Rigid-Body Dynamics" Vallery/Schwab "Advanced Dynamics", in 1st (black), 2nd (blue), or 3rd (red) edition additional material: Greenwood Advanced Dynamics, Cambridge University Press, 2006
Assessment	The main assessment of this course is a final on-campus exam, taken individually. The exam is taken digitally, in computer rooms. A small bonus can be obtained via an additional individual mid-term exam, which has the same form as the final exam. The final exam and the mid-term are organized on-campus. They are proctored and open-book, which means that students may bring their book (Vallery&Schwab and/or Greenwood, see above.) and 1 double-sided A4 sheet of hand-written notes. The A4 sheet must be written by the student him/herself and have the students name printed on it. The final grade G for the course is calculated from the midterm grade M and the exam grade E as follows: If E < 5.5, then the final grade G=E. If E >= 5.5, then the final grade G=0.9*E + 0.1*max(M,E). Three examples to clarify the calculation of the grades: 1. Student 1 obtained the grades: Mid-term M=7, final exam grade E=5. As E<5.5, the final grade G=E=5. 2. Student 2 did not participate in the mid-term, but obtained a final exam grade of E=8. The students final grade is G=E=8.

	3. Student 3 participated in the mid-term and obtained M=8, and a final exam grade E=5.5. The students final grade is G=0.9*E+0.1*M=5.8. The mid-term exam is only offered in Q1, and its grade is only valid for this academic year. The resit of the course contains only a digital final exam.
Enrolment / Application	See Dutch version
Department	3mE Department Biomechanical Engineering
Judgement	See Dutch version
Contact	Rigid-Body-Dynamics-3me@tudelft.nl. Please do not use lecturers' individual e-mail addresses.

WB2630 Toets 2	Continuum Mechanics	3

Responsible Instructor	Dr.ir. L.F.P. Noel
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hourse 2/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	Dutch

Course Contents

1) Linear 3D continuum theory
a. Deformation tensor in 3D, linear
b. Stretch in arbitrary direction
c. Stress vector and stress tensor in 3D, linear
d. Equilibrium in 3D
e. Principal stress directions, shear stress acting on a plane
f. Generalized Hookes law: isotropic, anisotropic, homogeneous, heterogeneous

2) Energy principles
a. PVW without deformations
b. PVW with deformation
c. Minimum of potential energy

3) Systems with n-dof
a. Introduction to a systematic way of solving n-dof problems. The discussion will be based on simple spring or truss problems. Aspects: DOFs, generalized deformations, generalized stresses, generalized constitutive equations, discrete continuity and equilibrium, system assembly, system matrix, boundary conditions, solution

4) Finite Element Method
a. Approximations based on minimum of potential energy:
i. Shape or interpolation functions
ii. Natural coordinate systems
iii. Displacement based finite elements
iv. Derivation of simple finite element
b. Mass matrix + eigenfrequencies
c. Topics required to use FEM:
i. Distributed loads
ii. Local coordinates
iii. Convergence, mesh density and mesh refinement
iv. Kinematic coupling
v. Element choice
vi. Different disciplines
vii. Frequently made mistakes

5) Introduction to plates
a. Introduction to plates (in-plane as well as out-of-plane)
b. Kinematic model for plates, deformations of the reference plane
c. Linear constitutive equations for laminated plates (ABD matrix)
d. Link with minimum of potential energy
e. FEM
NOT included are dynamic BC, e.g. Kirchhoff equivalent loads and loads at corners

Study Goals

Linear 3D continuum theory
The student is able to:
Recognize problems for which continuum theory is not applicable
Explain and use the governing equations in 3D: continuity, equilibrium/eqs. of motion
Determine for known displacement fields the deformations, and can identify the limitations
Calculate deformations and stretch in a specified direction
Use and explain a stress description in 3D and is able to determine a stress vector
Determine shear stress, normal stress and principle stress and directions, describe the practical relevance
Apply generalized Hookes law, describe its relevance and limitations

Energy principles
The student is able to:
Describe PVW and apply it to simple problems to derive equilibrium equations
Describe MPE and apply it to simple problems to build approximate solutions
Recognize when PVE and MPE can be applied correctly

Systems with n-dof:
The student is able to:
Describe the solution approach for simple and linear n-DOF problems from a global perspective.
Set up a set of equations manually for relatively simple and linear problems
Solve problems with few DOF entirely

FEM:
The student is able to:
Describe the basics of the finite element method
Describe the basic steps required for the formulation of a simple finite element
Perform a linear FEM analysis with a commercial FEM package (although training for a specific FEM package will be required)

Plates:
The student is able to:
Give a global description of plate/shell models
Recognize when plate/shell models are preferred.
Specify kinematic boundary conditions
Use min of potential energy to obtain simple approximations for deflections, stresses and deformations for laminated plates

Education Method

Contact hours:
Integrated plenary sessions, including lecture, exercises, and in-class tests (4 hours per week)
Weekly homework
Question hours (2 hours per week)

	In-class tests: In-class tests provide bonus credits Bonus credits are only valid for one academic year. Thus, the bonus credits evaporate at the end of the academic year During the in-class tests you may collaborate with your neighbor students and use internet info, slides, reader, books. Use of any type of electronic communication (sms, cell phone, email, facebook, whatsapp, etc etc.) is considered exam fraud and will be treated as such Grade calculation: Provided the grade of the written exam is 5.00 or higher: Final-Grade = 0.5 * (Grade-written-exam) + 0.5 * MAX(Grade-written-exam, Grade-inclass-tests) Grade-inclass-tests is based on average of best 70% of individual test grades. No participation in a test leads to a 0 as grade for that particular test. In case the grade of the written exam is lower than 5.00, one can not benefit from the bonus credits. Example: the grade of the written exam is 4.95, in that case there can be no benefit from the bonus credits Study materials: Reader (available in pdf format) Mechanics of Materials, Hibbeler, (Buckling Chp. 13, Energy Methods Chp. 14) Concepts and applications of finite element analysis, Robert Cook et al., Wiley could be used for further reading
Assessment	Write exam (bonus credits can be obtained with in-class tests)
Special Information	In the academic year 2017-2018 the lectures have been presented in English.
Department	3mE Department Precision & Microsystems Engineering

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

Maritieme Praktijk blok

MT2429-22	Maritime Markets and Operations	6
Responsible Instructor	Prof.ir. J.J. Hopman	
Instructor	Prof.dr.ir. E.B.H.J. van Hassel	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT2431	Geometrie en Stabiliteit 2	6
Responsible Instructor	Dr.ing. C.H. Thill	
Instructor	Ing. B. Goris	
Assistent	J.G. den Ouden	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	Please refer to Dutch page	
Department	3mE Department Maritime & Transport Technology	

MT2431 Toets 1	*Project Opdracht*	*3*

MT2431 Toets 2	*G&S2 Tentamen*	*3*

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

MT204-11 Maritieme Project blok

MT2434	2e Integratie Project	12
Responsible Instructor	Prof.ir. J.J. Hopman	
Responsible Instructor	Ir. N.C.H. Troost	
Instructor	E.H.M. Ulijn	
Instructor	Dr. P.E. Vermaas	
Instructor	Ing. B. Goris	
Instructor	Dr. E. Papadimitriou	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	Dutch	
Tags	Analysis Design Ethics Integrated Mechanics Modelling Optimalisation Process Project Small groups Sustainability	
Department	3mE Department Maritime & Transport Technology	

MT2434 Toets 1	*Project Opdracht(en)*	*10*

MT2434 Toets 2	Ethiek en Veiligheid tentamen	2

Responsible Instructor	Dr. E. Papadimitriou
Contact Hours / Week x/x/x/x	0/0/2/0
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	Dutch
Course Contents	Onderdeel 1 - Veiligheidskunde Het onderdeel veiligheidskunde is erop gericht een bewustzijn op te bouwen van de risicos in de dagelijkse praktijk en inzicht te geven in instrumenten om die risicos te identificeren en te beheersen. Als introductie tot de veiligheidskunde wordt ingegaan op: Bewustwording van gevaren en risicos, veiligheid als gevolg van menselijke, technische en organisatorische factoren, Methoden voor het systematisch analyseren van risicos. Aansluitend worden verschillende cases uitgewerkt, die in combinatie met Ethiek worden voorgelegd. Vanuit die lijn zijn de leerdoelen geformuleerd die, met op de discipline afgestemde cases, worden toegelicht. Onderdeel 2 - Ethiek Het ethiek onderdeel richt zich op de vaardigheden die nodig zijn om ethische en maatschappelijke problemen, waarmee ingenieurs in hun beroepspraktijk te maken kunnen krijgen, te herkennen en er verantwoordelijk mee om te gaan. Als introductie tot de Ethiek wordt ingegaan op: Het verwerven van (achtergrond) kennis en (cognitieve) vaardigheden die noodzakelijk zijn voor het herkennen, analyseren, en oplossingsgericht redeneren over, en bediscussiëren van ethische en maatschappelijke problemen waarmee ingenieurs in de beroepspraktijk te maken krijgen. Het in staat zijn tot zelfstandige gefundeerde oordeelsvorming over deze problemen en over mogelijke of voorgestelde oplossingsrichtingen. Het bevorderen van maatschappelijk verantwoordelijkheidsbesef en voorbereiding op een maatschappelijk verantwoorde beroepsuitoefening. Aansluitend hierop moet een uitgebreide case worden uitgewerkt, die in combinatie met Veiligheidskunde wordt voorgelegd. Vanuit deze insteek zijn de leerdoelen geformuleerd, die met op de discipline afgestemde cases worden toegelicht.
Study Goals	De student kan: (Veiligheidskunde) 1) de gangbare definities van gevaar en risico reproduceren en hun betekenis voor ontwerp en gebruik van technische systemen illustreren 2) enkele belangrijke veiligheidprincipes en -normen weergeven 3) onderkennen dat gevaren en risicos rond bestaande of nieuwe ontwerpen of systemen kunnen worden geïdentificeerd door gebruik te maken van enkele veelgebruikte analyse- of model technieken zoals het H-B-T (Hazard- Barrier-Target) model, de Veilige Envelop etc. 4) de interactie tussen mens en machine als bron van falen herkennen 5) risicos modelleren met decompositie-technieken (Foutenboom, Gebeurtenissenboom, het BowTie model) die ook ten grondslag liggen aan Kwantitatieve Risico Analyse 6) de problemen onderscheiden die verbonden zijn aan het afleiden van kwantitatieve gegevens (Ethiek) 7) de verschillende manieren herkennen waarop begrippen als verantwoordelijkheid en morele of maatschappelijke verantwoordelijkheid in het taalgebruik gehanteerd worden, en de verschillende opvattingen over de verantwoordelijkheid van ingenieurs 8) de mogelijke functies en beperkingen van beroepscodes verwoorden 9) de barrières herkennen die ingenieurs veelal ondervinden in de beroepspraktijk als ze verantwoord willen handelen 10) reflecteren op het vergroten van de ethische aanvaardbaarheid van risicos door ontwerpaanpassingen 11) argumenten formuleren en verdedigen ten aanzien van verantwoordelijkheid en aansprakelijkheid van actoren
Education Method	werkcolleges en project: deelname is verplicht
Assessment	De eindtoets bestaat uit een multiple choice tentamen. Indien er sprake is van onvoorziene omstandigheden, of maatregelen als gevolg van COVID-19, wordt als volgt afgeweken van de voorgeschreven toetsing: Remote opdracht. Het cijfer voor deze cursus is gebaseerd op de resultaten van huiswerkopdrachten voor Ethiek, respectievelijk voor Veiligheidskunde en op het tentamen. Een minimum score van 5.0 is vereist voor elke opdracht. Het eindcijfer is gebaseerd op de volgende indeling: 25% VK-opdrachten, 25% Ethiek-opdrachten en 50% tentamen.
Department	3mE Department Maritime & Transport Technology

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

3e jaar BSc MT 2022

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

3e jaar BSc MT minor 2022

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Bachelor Marine Technology**

3e jaar BSc MT major 2022

MT3420	Elektrische Systemen & Regeltechniek	6

Responsible Instructor	Dr.ir. H. Polinder
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	Dutch
Required for	MSc Maritime Technology
Course Contents	- Introduction to drive and energy systems - Electricity and electrical circuits - Magnetism and magnetic circuits - Power electronics - DC machines - AC voltages, currents, powers and phasors - Transformers - Synchronous machines (including PM machines) - Induction machines - Choices in electrical drive and energy systems - Introduction to control - Dynamic models and transfer functions - Dynamic response - Analysis of feedback control - Frequency response
Study Goals	- Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws. - Sketch phasor diagrams with voltage and current phasors. - Calculate active, reactive and apparent power in AC circuits. - Calculate energy stored in capacitors and inductors. - Describe the most important power semiconductors (diodes, IGBTs, MOSFETs) and their switching behaviour. - Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs). - Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation). - Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode. - Calculate induced voltage using Faradays law. - Calculate electromagnetic forces and torques using Lorentz law or the power balance. - Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines. - Derive performance characteristics based on these equivalent circuits (torque-speed characteristics). - Describe advantages and disadvantages of electrification of drives. - Select suitable machine and drives for different applications. - Recognize open-loop and closed-loop control systems. - Describe elements in the control loop. - Describe control systems using block diagrams. - Model and linearize simple physical systems (in particular mechanical, electrical, electromechanical, fluid and thermodynamic systems). - Derive transfer functions from simple linear differential equations, using Laplace transforms. - Analyse the dynamic behaviour of closed-loop systems based on transfer functions in terms of step and frequency response or of poles and zeros. - Set up PID controllers for simple systems and apply fine-tuning rules (such as Ziegler & Nichols). - Interpret Bode diagrams for 1st and 2nd order systems. - Apply the aforementioned control methods to electrical machines by modelling an electrical machine and drawing up a transfer function for it.
Education Method	Lectures
Books	T. Wildi, "Electrical Machines, Drives and Power Systems" G.F. Franklin, J.D. Powell, A. Emami-Naeini, Feedback Control of Dynamic Systems
Assessment	The examination will be as follows: written exam, closed book + obligatory practical. Students pass the course if the result of the exam is >=5,8 and students passed the practical, and is equal to the the result of the exam. The result of the practical is only valid in the year that the practical is done. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: written exam, open book, online.
Department	3mE Department Maritime & Transport Technology

MT3432	Organisatie van Scheepsproductie	6

Responsible Instructor	Ir. M. van der Schaaf
Responsible Instructor	Dr.ir. J.F.J. Pruijn
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	Dutch
Department	3mE Department Maritime & Transport Technology

MT3432 Toets 1	*OVS Tentamen*	*2.5*

MT3432 Toets 2	*Report*	*3.5*

MT3433	Complexe Belastingen en Trillingen	6
Responsible Instructor	Dr. A. Grammatikopoulos	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	This course requires mastery of the following subjects: Matrix algebra (WBMT1051: Wiskunde2) Eigenvalues and Eigenvectors (WBMT1051: Wiskunde 2) Partial differential equations (WBMT2048: Wiskunde 3) Newton's Laws (WB1135: Introductory Dynamics) Dynamics of rigid bodies (WB2630: Advanced Mechanics) Tension, bending, torsion, elongation, shear (WB1631-15: Sterkteleer) Complex numbers, trigonometric functions, logarithms Practical use of Python (alternatively MATLAB or Maple, but support will only be provided for Python)	
Course Contents	1) General knowledge: Phasor notation Fourier transform (frequency content of time signals) Fluid effects on vibration (added mass/damping/stiffness) 2) Vibration of discrete systems with 1 or more degrees of freedom: Equations of Motion (free body diagram, Lagrange method) Free response (natural frequencies, mode shapes, damping) Forced response (transfer functions) 3)Vibration of continuous systems: Analytical methods (natural frequencies, mode shapes) Approximation methods System coupling 4) Structural waves and noise: Mechanical waves Noise fields and sound paths Transmission, reflection, absorption, radiation Decibel scale and frequency bands Noise propagation on ships 5) Application: Sea-induced vibration Machinery-induced and propeller-induced vibration Countermeasures Experiments & measurements	
Study Goals	The student can predict, measure and assess the structural dynamic behaviour (noise and vibration levels) of complex discrete and continuous mechanical systems. Specifically, this means that the student: Can determine the dynamic behaviour of 1DOF systems, 2DOF systems and simple continuous systems analytically. Can use numerical methods within a programming environment, such as Python, to solve more complex systems, for which an analytical solution does not exist. Can apply the modal summation principle to MDOF systems and continuous systems. Can describe the main sources, transmission paths and analysis methods for noise and vibration. Can suggest countermeasures to reduce noise and vibration.	
Education Method	The material will be delivered through lectures and tutorials. In the event of unforeseen circumstances or measures as a result of COVID-19, these lectures and tutorials may take place online. In order to process the lecture material and to prepare for the exam, the student is expected to individually complete exercises every week. Some of the exercises will involve development of code during the tutorials and others will be in the form of online quizzes. The code developed within the tutorials will be used to perform the practicum calculations. The aspects of analytical and numerical prediction, measurement and analysis of the dynamic behaviour of a mechanical system will treated in a practicum in small groups. In the event of unforeseen circumstances or measures as a result of COVID-19, it is possible that the measurements of the practicum may be cancelled. In that case, only the analytical and numerical calculations will be performed and reported. The total expected time needed to complete the course is 168 hours per student, with the following breakdown: 32 hours of lectures (4 hours per week) 16 hours of tutorials (2 hours per week) 16 hours for weekly practice quizzes (2 hours per week) 29 hours for the practicum (5 hours for initial planning and analytical calculations, 15 hours for numerical calculations, 3 hours of measurements, 9 hours for analysis and reporting) 32 hours for self-study (4 hours per week during the semester, can be partly used to further support other aspects, such as the weekly quizzes or the practicum) 40 hours of preparation for the exam 3 hours exam	
Reader	The MT3402 reader Scheepstrillingen en geluid is used for this course.	
Assessment	The assessment takes place in 2 parts, each of which must be completed with a pass: Exam (summative) Practical report (formative) If possible, the examination will be held physically in the form of a written examination. In the event of unforeseen circumstances or measures due to COVID-19, the assessment will take place online using MapleTA/Mobius.	
Permitted Materials during Tests	The following materials are allowed during the exam: Calculator (non-programmable) Ruler or triangle Blue or black pen Any material in hard copy (printed or handwritten)	
Remarks	You must register to participate in this course. Registration for a course is not the same as registration for a (partial) exam, registration in Brightspace or registration for a study in Studielink.	
Department	3mE Department Maritime & Transport Technology	

MT3BEP	Bachelor End Project	12
Responsible Instructor	Dr.ir. H.J. de Koning Gans	
Instructor	Dr.ir. H.J. de Koning Gans	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	Het Bachelor Eind Project (BEP) is het laatste onderdeel van de bachelor Maritieme techniek (MT), heeft een omvang van 12 ECTS en wordt uitgevoerd in periode 1 en 2 of periode 3 en 4. Het project fungeert als capstone, een afsluitend en samenvattend project voor de gehele bachelor. Het project wordt uitgevoerd in groepen van 4 studenten. De opdracht van het project is voor iedere groep studenten verschillend maar heeft in alle gevallen betrekking op een multidisciplinaire (ontwerp)opdracht waarin een duidelijke onderzoekscomponent aanwezig is. Deze (ontwerp)opdracht kan betrekking hebben op een heel schip, een deel van een schip of een aan scheepsbouw of scheepvaart gerelateerd proces. Opdrachten worden geformuleerd door of namens U(H)Ds en hoogleraren van MT. De opdracht bevat zowel een verdiepende onderzoekscomponent als een multidisciplinaire ontwerp en/of analyse component, maar de balans tussen beiden kan per opdracht binnen redelijke grenzen vrij gekozen worden. Aan beide componenten wordt tenminste ruwweg 25% van de tijd besteed. De aard van de opdracht is altijd dusdanig dat het de studenten noodzaakt om tenminste 2 disciplines binnen MT te combineren.	
Department	3mE Department Maritime & Transport Technology	

Dr.ir. J.H. den Besten

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Prof.dr.ir. E.B.H.J. van Hassel

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Dr.ir. H.J. de Koning Gans

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Department	Ship Hydromechanics and Structures
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Dr.ir. M. Langelaar

Unit	Mech, Maritime & Materials Eng
Department	Structural Optimization and Mechanics
Telephone	+31 15 27 86506
Room	34.G-1-300

Ir. A.M. van der Niet

Department	Support Cognitive Robotics
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Ir. A.A. Nijdam

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Dr.ir. L.F.P. Noel

Department	Computational Design and Mechanics
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J.G. den Ouden

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics and Structures
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Ir. D. den Ouden-van der Horst

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
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Dr. E. Papadimitriou

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
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Dr. R. Pecnik

Department	Energy Technology
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Dr.ir. H. Polinder

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Department	Transport Engineering and Logistics
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Dr.ir. J.F.J. Pruijn

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